CN7026 Cloud Computing

Vertical Sector Smart Campus

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Vertical Sector Smart Campus Introduction

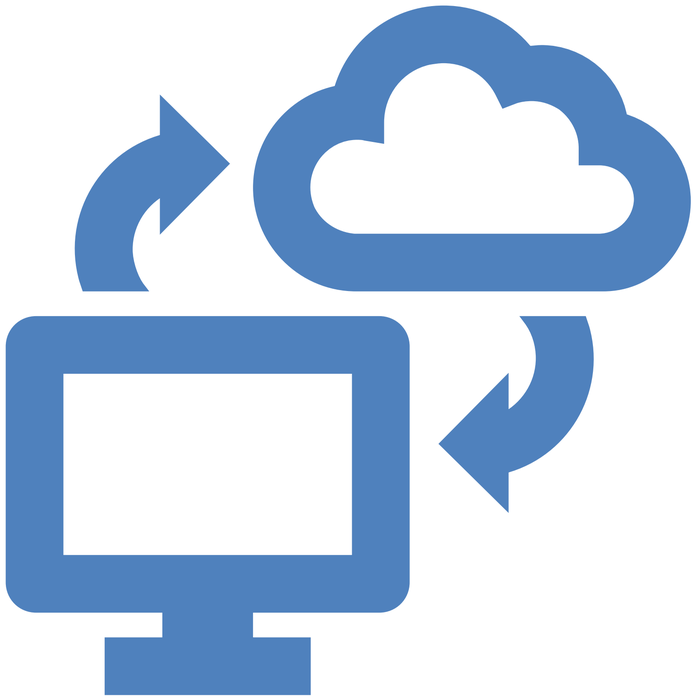
- **Transformation: Institutions are becoming smart learning environments with IoT, AI, and data analytics.**
- **Problem: Data decentralization causes inefficiency, slow decision-making, and poor tracking (e.g., attendance).**
- **Solution: Cloud computing can connect systems and enable real-time data analysis.**
- **Smart Campus Hub: Cloud-based platform to manage resources and improve student experiences.**
- **Benefits: Centralized data helps administrators make better decisions, boosts sustainability, and enhances digital learning.**

Project Plan

Week	Task	Milestone
Week 1-2	Requirement analysis, stakeholder consultation, and defining key objectives	Project scope finalised
Week 3-4	Infrastructure assessment, identifying existing campus systems, and selecting cloud service providers	Cloud platform selection completed
Week 5-6	Designing cloud-based Smart Campus Hub architecture, ensuring scalability and integration	System architecture blueprint ready
Week 7-8	Developing data integration pipelines for real-time analytics and cross-departmental access	Data integration framework implemented
Week 9-10	Implementing security measures, access control, and compliance standards	Security protocols established
Week 11-12	Testing system functionality, performance evaluation, and user feedback collection	System validation completed
Week 13-14	Training staff, onboarding students, and deploying smart applications	Full system deployment
Week 15	Monitoring, troubleshooting, and optimizing for efficiency	Project completion and review

Functional Requirements :-

- **AI Analytics**: Real-time data for decision-making.
- **Centralized Data**: Unified storage across departments.
- **Automated Scheduling**: Smart classroom and utility management.
- **IoT Integration**: Sensor-based security and energy monitoring.
- **Cloud Learning**: Seamless access to digital resources.
- **Smart Attendance**: Facial recognition or RFID tracking.
- **Predictive Maintenance**: AI alerts for equipment issues.
- **Mobile/Web Access**: Unified platform for all users.
- **Incident Detection**: Automated alerts for emergencies.
- **Dynamic Access**: Role-based authentication and permissions.

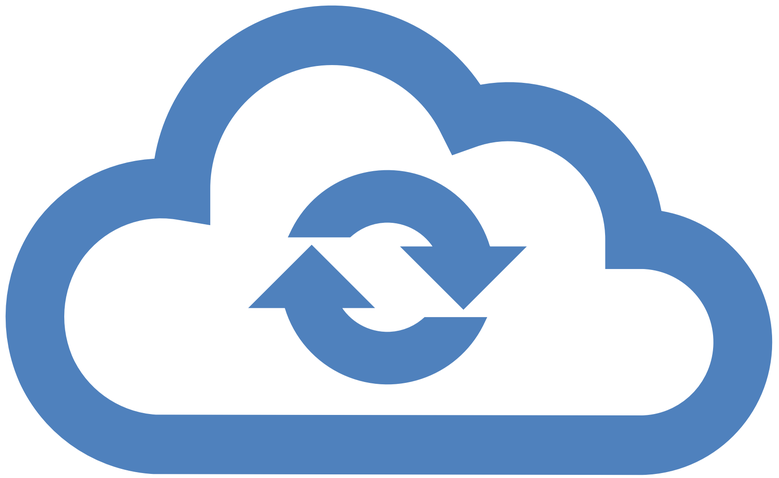


Non-Functional Requirements

- **Scalability**
  - ❖ Handles growing users, data, and devices without performance issues.
- **Low Latency**
  - ❖ Real-time responses for critical applications like security alerts and AI analytics.
- **Regulatory Compliance**
  - ❖ Adheres to **GDPR** and **FERPA** for student data privacy and protection.
- **User-Friendly Interface**
  - ❖ Easy-to-use dashboards and seamless navigation for all users (students, faculty, admins).

Data Centre

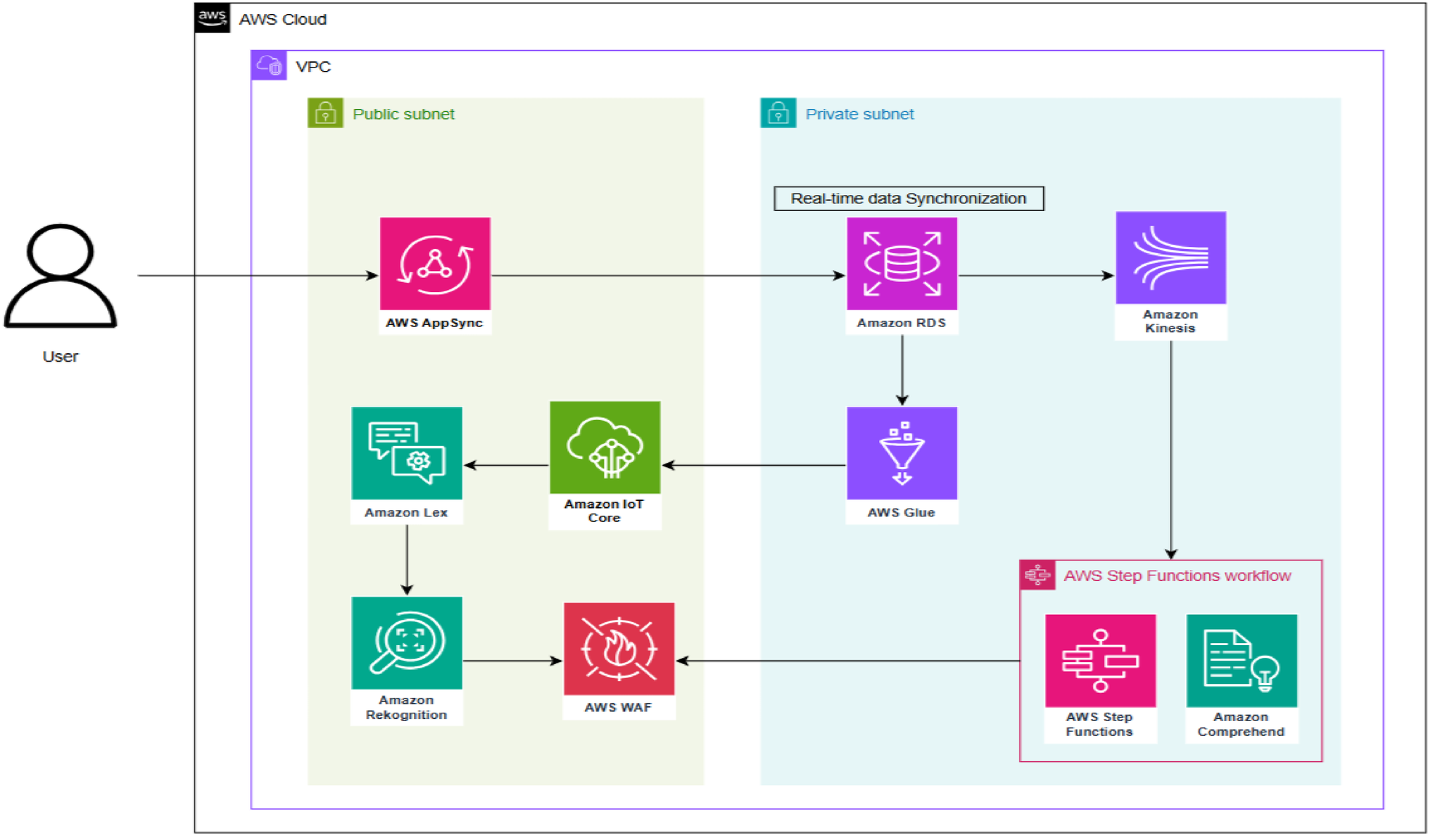
- **Location Benefits**:
  - ❖ Advanced digital infrastructure.
  - ❖ Strong data protection laws (GDPR compliance).
  - ❖ Proximity to educational institutions.
- **Security & Compliance**:
  - ❖ Compliant with **ISO/IEC 27001** for information security.
  - ❖ Uptime Institute Tier III or Tier IV for cloud service reliability.
  - ❖ **Cyber Essentials** certification for enhanced cybersecurity.
  - ❖ **SOC 2** compliance for data security and privacy.
  - ❖ **BS 10012** for personal data governance.
- **Conclusion**:
  - ❖ London offers a secure, scalable, and compliant cloud infrastructure for smart campuses.



Cloud Architecture Solution (AWS)

- **Why AWS**:
  - Supports AI, scalability, and digital learning enhancements.
- **Key AWS Services**:
  - **AWS Lambda** & **Amazon Sage Maker**: Real-time data analysis for resource management and maintenance scheduling.
  - **Amazon S3**: Stores learning materials and teaching content.
  - **AWS CloudFront**: Fast global delivery of learning resources.
  - **AWS IAM**: Secure, role-based access management for users.
- **Smart Systems Integration**:
  - AI-based facial recognition for **smart attendance** and security.
  - IoT-enabled automation for improved campus efficiency.
- **Outcome**:
  - Increased efficiency, enhanced learning experiences, and greater student engagement.

Cloud Architecture



Costing

AWS Pricing Calculator > My Estimate

My Estimate Edit

Export

Estimate date: March 17, 2025

Estimate summary Info

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	8,002.11 USD	96,025.32 USD Includes upfront cost

Getting Started with AWS

Get started for free

Contact Sales

Service Name	Status	Upfron...	Monthly...	Descrip...	Region	Config Summary
Amazon Simple Storage Service (S3)	-	0.00 USD	4.80 USD	-	Europe (London)	S3 Standard storage (2...
AWS IoT Core	-	0.00 USD	42.05 USD	-	Europe (London)	Number of devices (MQ...
Amazon RDS for Db2	-	0.00 USD	7,223.56 USD	-	Europe (London)	Storage amount (20 GB...
AWS AppSync	-	0.00 USD	40.00 USD	-	Europe (London)	Number of API Request...
Amazon Kinesis Data Streams	-	0.00 USD	156.80 USD	-	Europe (London)	Duration of data retenti...
AWS Web Application Firewall (WAF)	-	0.00 USD	500.00 USD	-	Europe (London)	Number of Web Access ...
AWS Step Functions	-	0.00 USD	34.90 USD	-	Europe (London)	Workflow requests (700...
Amazon Comprehend	-	0.00 USD	0.00 USD	-	Europe (London)	Number of documents ...

THANK YOU !!!

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